PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Self-testing for pregnancy: a systematic review and meta-analysis
AUTHORS	Kennedy, Caitlin; Yeh, Ping; Gholbzouri, Karima; Narasimhan, Manjulaa

VERSION 1 – REVIEW

REVIEWER	Julie Shaw Eastern Ontario Regional Laboratories Association
REVIEW RETURNED	05-Aug-2021

GENERAL COMMENTS	It is unfortunate that few studies were available for the author's review and that none of the studies included a cost analysis for the use of home/self pregnancy testing versus pregnancy testing through the healthcare system alone. This speaks to the need for studies on this topic, particularly in lower income settings and in settings where women would benefit from greater accessibility. I have only one suggested edit, which is in the introduction. The authors state that urine pregnancy tests are 100% sensitive and specific. I would not quote 100% for either sensitivity or specificity as this is misleading. We know that there are false positives and false negatives, dependent upon the form of hCG measured by the pregnancy test in question. It will also depend on the user, training etc. You could say that the sensitivity and specificity are close to 100%.

REVIEWER	Christina C. Mehta Emory University School of Medicine, Medicine, Division of Infectious Diseases
REVIEW RETURNED	10-Aug-2021

GENERAL COMMENTS	This is a really well-conducted and well-described systematic review on pregnancy self-testing compared to clinic testing. Just a few minor suggestions to help improve clarity of the paper: -The rationale for comparing self-testing to facility testing could be expanded, especially in light of the few papers that were reviewed. Is the assumption that all women have access to pregnancy testing through a clinic? - Although mentioned in the limitations, the limits to generalizability due to conducting most of the studies in populations that experienced abortions could be further described. This is a major limitation. -According to this review, there is no information on some the outcomes of interest (such as for efficacy: missed ongoing pregnancy, self-efficacy/self-determination, mental health, adverse events, device-related issues). Would highlight this in the discussion as a call for future research. -Suggest moving the sections relevant to all reviews (such as:

	search strategy, search terms, screening abstracts, data extraction and management) to before the effectiveness review in the methods. Currently, it is a little confusing what methods carry over to the values and preferences and cost reviews. -For values and preferences review, outcomes of interest beyond preference for self-testing were ability to access, informed decision-making, coercion, confidentiality, self-determination, health decision-making, and seeking redress. Would appreciate a summary paragraph that describes relevant findings for these outcomes in the results section. -Is appropriate clinical follow-up the same concept as loss to follow-up? I'm not sure how to interpret this finding in light of the population studied (post-abortion) as the consequences would be different than for pre-natal follow-up. Think this is an assumption that could be further discussed in limitations. -Additional information on test effectiveness score (page 4, line 20) would be helpful to interpreting the results presented.

REVIEWER	Karen Cowgill University of Washington
REVIEW RETURNED	10-Aug-2021

GENERAL COMMENTS	The editors requested that I perform a statistical review of this manuscript, and so my comments are limited to the meta-analysis shown in Figure 2 and reported in the text. My recommendation is that the meta-analysis be omitted from the manuscript and that those 4 studies be reviewed qualitatively along with the others in the systematic review. The authors acknowledge that with an I2 of 87%, the heterogeneity is high. They have done well to use a random-effects model and to make an effort to conduct a subgroup analysis of HIC vs LMIC settings, but in my opinion these techniques are not sufficient to overcome the small number of studies (k = 4) and the high heterogeneity. While the outcome measured in these studies may be sufficiently similar to group them, it's likely that the differences in the patient populations and settings as well as in the study methodology are too great for the summary RR to be meaningful. Better to explore those differences than to try to create a summary measure that may be misleading. Non-statistical side note: interesting topic, and surprising there's not more published about it!

REVIEWER	Katharine Correia Amherst College, Mathematics & Statistics
REVIEW RETURNED	17-Aug-2021

GENERAL COMMENTS	This paper provides an interesting review of the literature-to-date on the effectiveness, values and preferences, and cost of pregnancy self-testing. The objectives are explicitly stated, and the methods are explained clearly such that the review could be reproduced. Although the initial goals were quite broad, addressing effectiveness, values and preferences, and cost – and stratifying analyses in a number of different ways -- the identified studies only allowed for a portion of these intended analyses to be conducted (suggesting more research should be done in certain areas). I only have minor comments: 1. Pg 8, line 23: Should there be a citation for the Comprehensive Meta-Analysis software used? 2. Pg 8, lines 50-53: Clarity may be improved if this is broken into

	more than one sentence; there are a lot of i.e.'s, e.g.'s, and etc.'s in the parentheses 3. Pg 9, lines 11-19: It is a little confusing the way the "Cost review" methods text states "Cost literature was summarized qualitatively. Cost literature was classified into four categories …" but then no cost studies were identified to be included in review. Maybe re-state so it's clear that was the intention/plan, but not actually completed (due to no relevant articles). 4. Figure 1: Please double-check the numbers in Figure 1: number of full text articles retrieved to determine eligibility is 62. Number excluded is 46, and number included is 20 (6 PICO + 18 in VP – 4 in both = 20). 46+20 = 66? Also, the reasons "no cost data" and "no values and preferences data" is a little unclear (why were these deemed eligible originally if not related)? 5. Table 3: In the Harper et al. row, last column: should while/black be white/black? 6. Pg 19, lines 9 – 11: Why were studies excluded if they compared groups of individuals given home pregnancy test to those given nothing or the standard of care? It seems like comparing to standard of care would have been a valuable comparison to consider. 7. Pg 19, lines 45-49: The paragraph starts with "This review had strengths and limitations", but seems to only focus briefly on limitations. Do you want to emphasize some of its strengths (exhaustive search of multiple databases, articles double-data-entered, etc.)? The limitations could also be expanded upon. In particular, with regards to the limitation that the literature is heavily skewed on post-abortion care (which "represent only a fraction of individuals using pregnancy self-tests globally"), could you please add a few sentences describing the implications of this limitation? This seems to be a pretty big limitation, but is a limitation of the literature in this area, rather than a review-level limitation. For instance, this could be a chance to highlight implications for future research. 8. Figure 2: Figure 2 is a little blurry; a p-value can't be 0, please replace with <0.0001

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1
Dr. Julie Shaw, Eastern Ontario Regional Laboratories Association

Comments to the Author:
It is unfortunate that few studies were available for the author's review and that none of the studies included a cost analysis for the use of home/self pregnancy testing versus pregnancy testing through the healthcare system alone. This speaks to the need for studies on this topic, particularly in lower income settings and in settings where women would benefit from greater accessibility.

Thank you. We hope this review and the attention from the WHO guideline may spur some of this research.

I have only one suggested edit, which is in the introduction. The authors state that urine pregnancy tests are 100% sensitive and specific. I would not quote 100% for either sensitivity or specificity as this is misleading. We know that there are false positives and false negatives, dependent upon the form of hCG measured by the pregnancy test in question. It will also depend on the user, training etc. You could say that the sensitivity and specificity are close to 100%.

Thank you for your positive comments. We have made this change to the introduction. The sentence now reads: "The current generation of tests have close to 100% sensitivity and specificity in detecting hCG at concentrations of 25 milli-IU/ml or more, and thus are able to detect pregnancy as early as one day after a missed menstrual period.[1]"

Reviewer: 2
Dr. C. Christina Mehta, Rollins School of Public Health

Comments to the Author:
This is a really well-conducted and well-described systematic review on pregnancy self-testing compared to clinic testing. Just a few minor suggestions to help improve clarity of the paper:

Thank you.

-The rationale for comparing self-testing to facility testing could be expanded, especially in light of the few papers that were reviewed. Is the assumption that all women have access to pregnancy testing through a clinic?

The review was designed through a self-care lens, in the liminal space of actions that end users take to promote or monitor their own health (e.g. self-testing for pregnancy) compared with actions that providers or health systems take on behalf of the end user. For this reason, we framed the review question stated at the beginning of the Methods section as: should self-testing for pregnancy be available as an additional option to health facility-based testing?

- Although mentioned in the limitations, the limits to generalizability due to conducting most of the studies in populations that experienced abortions could be further described. This is a major limitation.

We have added this limitation to the conclusions section of the abstract through the addition of this new text: “Effectiveness data is mostly available from articles on post-abortion care”. We have also edited the discussion section where we discuss the effectiveness review findings to particularly highlight this issue of generalizability. The revised section now reads: “The fact that almost all data came from post-abortion care settings limits the generalizability of our findings and the conclusions that can be drawn about public health benefits of this intervention.”

-According to this review, there is no information on some the outcomes of interest (such as for efficacy: missed ongoing pregnancy, self-efficacy/self-determination, mental health, adverse events, device-related issues). Would highlight this in the discussion as a call for future research.

We have added this to the discussion section. The newly added text now reads as follows:

“Further research on the effectiveness (especially on outcomes for which we found no comparative data, i.e. missed ongoing pregnancy; self-efficacy, self-determination, autonomy, and empowerment; mental health and well-being; adverse events and social harms; and device-related issues), values and preferences, and costs of pregnancy self-testing is needed particularly in low-resource settings and in contexts where people would benefit from greater accessibility.”

-Suggest moving the sections relevant to all reviews (such as: search strategy, search terms, screening abstracts, data extraction and management) to before the effectiveness review in the methods. Currently, it is a little confusing what methods carry over to the values and preferences and cost reviews.

Thank you for this suggestion. We have reviewed the methods section for clarity with this comment in mind. Where we outline the methods for the complementary values and preferences and cost reviews, we have now clarified that we used the same search terms and screening process before we list the specific inclusion criteria and data extraction/management plans.

-For values and preferences review, outcomes of interest beyond preference for self-testing were ability to access, informed decision-making, coercion, confidentiality, self-determination, health decision-making, and seeking redress. Would appreciate a summary paragraph that describes relevant findings for these outcomes in the results section.

Thank you for this excellent comment. When we were designing the values and preferences review, we wanted to be as inclusive as possible on potential end-user and health worker perspectives on

pregnancy self-testing – so we included outcomes from a human rights lens including ability to access, informed decision-making, coercion, confidentiality, self-determination, health decision-making, and seeking redress. We have now added some additional text to the results section on these points, synthesizing findings beyond simply preference, as follows:

“While most values and preferences studies focused on acceptability, ease of use, and reasons for preferring pregnancy self-testing over other options, a few studies also presented findings on our other areas of interest: ability to access, informed decision-making, coercion, confidentiality, self-determination, health decision-making, and seeking redress. One study from the USA found themes of privacy, empowerment, consumer choice, and meaning-making among women using pregnancy tests [22]. Several qualitative studies from the USA and Scotland noted that pregnancy tests shape individuals’ relationships with their bodies and their social roles, relationships, and responsibilities [30-31]; one theorized that testing is seen as socially and morally mandatory as “the cultural script produces an assumption, impetus, and moral obligation that women must test” to know what is inside their bodies and how it may shape their lives [30]. Finally, one study from India found that managing abortion at home meant that women were able to avoid the inconvenience of travel, childcare, and housework, as well as maintain confidentiality [28]. The pregnancy test alleviated women’s anxieties: they found it reassuring, and several experienced a sense of relief.”

-Is appropriate clinical follow-up the same concept as loss to follow-up? I'm not sure how to interpret this finding in light of the population studied (post-abortion) as the consequences would be different than for pre-natal follow-up. Think this is an assumption that could be further discussed in limitations.

The PICO outcome "appropriate clinical follow-up" was designed to be purposefully broad, including counseling, antenatal care visit(s), contraceptive services, abortion services, etc. In the post-abortion context, we did not find reported quantitative comparative outcomes on these other aspects of follow-up but did find "loss to follow-up". In these four RCTs, loss to follow-up was defined as when the client did not return for their follow-up visit or was not able to be contacted by phone – after the abortion and self-testing for pregnancy. In our results section, we separated this finding from the "appropriate clinical follow-up" outcome of number of clients who received antenatal counseling at their CHW visit, which was another measure of this broad outcome.

-Additional information on test effectiveness score (page 4, line 20) would be helpful to interpreting the results presented.

Thank you for this suggestion. We have added a definition of "test effectiveness score" as used by the meta-analysis we cite in the introduction: "discriminatory power, where a score of 1.0 or less indicates poor distinctions between pregnant and nonpregnant individuals, and higher scores imply greater effectiveness".

Reviewer: 3
Dr. Karen Cowgill, University of Washington

Comments to the Author:
The editors requested that I perform a statistical review of this manuscript, and so my comments are limited to the meta-analysis shown in Figure 2 and reported in the text.

My recommendation is that the meta-analysis be omitted from the manuscript and that those 4 studies be reviewed qualitatively along with the others in the systematic review. The authors acknowledge that with an I2 of 87%, the heterogeneity is high. They have done well to use a random-effects model and to make an effort to conduct a subgroup analysis of HIC vs LMIC settings, but in my opinion these techniques are not sufficient to overcome the small number of studies (k = 4) and the high heterogeneity. While the outcome measured in these studies may be sufficiently similar to group them, it’s likely that the differences in the patient populations and settings as well as in the study methodology are too great for the summary RR to be meaningful. Better to explore those differences than to try to create a summary measure that may be misleading.

Non-statistical side note: interesting topic, and surprising there’s not more published about it!

Thank you – to both the editors and this reviewer – for taking the time to specifically look at the statistical analysis portion of our manuscript. As stated above, we have decided to retain the meta-analysis as this was a crucial piece of evidence used in the final WHO guideline (we have now added the final WHO recommendation for this topic area to the discussion section). We have also added in a bit more details on qualitative synthesis of the articles which were included in the meta-analysis (although this was limited due to space constraints).

Reviewer: 4
Dr. Katharine Correia, Amherst College

Comments to the Author:
This paper provides an interesting review of the literature-to-date on the effectiveness, values and preferences, and cost of pregnancy self-testing. The objectives are explicitly stated, and the methods are explained clearly such that the review could be reproduced. Although the initial goals were quite broad, addressing effectiveness, values and preferences, and cost – and stratifying analyses in a number of different ways -- the identified studies only allowed for a portion of these intended analyses to be conducted (suggesting more research should be done in certain areas). I only have minor comments:

1. Pg 8, line 23: Should there be a citation for the Comprehensive Meta-Analysis software used?

We have added a citation for CMA: Reference 13. Borenstein M, Hedges L, Higgins J, Rothstein H. Comprehensive Meta-Analysis Version 3. Englewood, NJ: Biostat; 2013.

2. Pg 8, lines 50-53: Clarity may be improved if this is broken into more than one sentence; there are a lot of i.e.'s, e.g.'s, and etc.'s in the parentheses

We have adjusted this parenthetical remark for clarity. It now reads: (by age, gender, or other factors. These could include legal restrictions around who can access, e.g., by age, requirement to be married to purchase, etc. These could also include practical barriers, e.g., is it harder for adolescents to access self-tests because they are not allowed out on their own, they have no money, etc.)

3. Pg 9, lines 11-19: It is a little confusing the way the "Cost review" methods text states "Cost literature was summarized qualitatively. Cost literature was classified into four categories …" but then no cost studies were identified to be included in review. Maybe re-state so it's clear that was the intention/plan, but not actually completed (due to no relevant articles).

Thank you – we have reworded this section to say this was the intention, but we did not actually do this portion of the methods because we found no includable articles for the complementary cost review. This section now reads:

The same search terms and screening process were used to identify studies to be included in the cost review. Studies would have been included in this review if they presented primary data comparing costing, cost-effectiveness, cost-utility, or cost-benefit of the intervention and comparison listed in the PICOs above, or if they presented cost-effectiveness of the intervention as it relates to the PICO outcomes listed above. If cost literature had been found, we would have summarized it qualitatively. We planned to classify cost literature into four categories (health sector costs, other sector costs, patient/family costs, and productivity impacts) and within each category would have organized by study design/methodology, location, and population.

4. Figure 1: Please double-check the numbers in Figure 1: number of full text articles retrieved to determine eligibility is 62. Number excluded is 46, and number included is 20 (6 PICO + 18 in VP – 4 in both = 20). 46+20 = 66? Also, the reasons "no cost data" and "no values and preferences data" is a little unclear (why were these deemed eligible originally if not related)?

Thank you for this catch! We have updated the PRISMA flowchart with more specific reasons (to better capture the complexity of doing three reviews in one search). The numbers now match up.

5. Table 3: In the Harper et al. row, last column: should while/black be white/black?

Thanks for the catch – you are correct. We have fixed this.

6. Pg 19, lines 9 – 11: Why were studies excluded if they compared groups of individuals given home pregnancy test to those given nothing or the standard of care? It seems like comparing to standard of care would have been a valuable comparison to consider.

As mentioned above, we were asked by the WHO to frame this review around using pregnancy self-testing as "an additional option" beyond facility-based testing, with the idea to expand WHO guidelines (which typically focus on the health sector) to self-care interventions.

7. Pg 19, lines 45-49: The paragraph starts with “This review had strengths and limitations”, but seems to only focus briefly on limitations. Do you want to emphasize some of its strengths (exhaustive search of multiple databases, articles double-data-entered, etc.)? The limitations could also be expanded upon. In particular, with regards to the limitation that the literature is heavily skewed on post-abortion care (which “represent only a fraction of individuals using pregnancy self-tests globally”), could you please add a few sentences describing the implications of this limitation? This seems to be a pretty big limitation, but is a limitation of the literature in this area, rather than a review-level limitation. For instance, this could be a chance to highlight implications for future research.

8. Figure 2: Figure 2 is a little blurry; a p-value can’t be 0, please replace with <0.0001

Thanks for this careful attention to our figures. We have re-done figure 2 so it is higher resolution and have replaced the value of 0 (the default of our meta-analysis software program) with <0.001.

VERSION 2 – REVIEW

REVIEWER	Julie Shaw Eastern Ontario Regional Laboratories Association
REVIEW RETURNED	04-Jan-2022

GENERAL COMMENTS	I still have concerns about the sensitivity and specificity information cited in the introduction for home pregnancy tests. I am asking the authors to address the fact that not all tests detect the same forms of B-hCG, which will alter sensitivity and specificity. There are some good papers in the literature on this issue. (Gronowski 2009, Nerenz 2014). It is also important to address that tests can be highly sensitive if the tests are performed as per the manufacturer's instructions. This is an important point given these are performed outside of the lab.

REVIEWER	Christina C. Mehta Emory University School of Medicine, Medicine, Division of Infectious Diseases
REVIEW RETURNED	07-Jan-2022

GENERAL COMMENTS	The revised manuscript adequately addresses all previous comments/questions.

REVIEWER	Karen Cowgill University of Washington
REVIEW RETURNED	04-Jan-2022

GENERAL COMMENTS	Thank you for the detailed response to my and the editors' and other reviewers' comments.

	It is helpful to understand that this is one of a series of analyses you conducted on commission from WHO, and I wonder whether including more information about the WHO guidelines and the process for developing them in the Methods would provide better context. My initial thought on reading your response was to reluctantly agree to leaving the meta-analysis in the article, but it still doesn't sit well with me. The concerns I have are: 1. Does the result of the meta-analysis add anything important to the GRADE table -- would you have arrived at a different recommendation if you hadn't done the meta-analysis? 2. I briefly reviewed the Annex to the WHO guidelines with the methods from WHO and GRADE and didn't see a requirement to perform a meta-analysis in either (perhaps I missed it) -- could you explain your thinking about what it adds? 3. How would removing the meta-analysis affect the overall message and evidence of this article? 4. Computing power makes it easy to plug in numbers and produce quantitative results that appear authoritative but may not be. 5. As an aside -- because I still think there are too few studies and too much heterogeneity to justify the meta-analysis -- you write that "Stratification by high- vs. low- and middle-income settings did not yield meaningful differences; no further stratifications were available." However, it seems that income category is just one of several potential variables you might stratify on. If the meta-analysis stays in the article, I suggest adding more disclaimers and qualifiers about its applicability and interpretation than the brief sentences in the revised version.

VERSION 2 – AUTHOR RESPONSE

Reviewer: 3
Dr. Karen Cowgill, University of Washington Comments to the Author:
Thank you for the detailed response to my and the editors' and other reviewers' comments.

Thank you for your continued helpful comments in making this manuscript as strong as it can be.

It is helpful to understand that this is one of a series of analyses you conducted on commission from WHO, and I wonder whether including more information about the WHO guidelines and the process for developing them in the Methods would provide better context.

We have now added in the first paragraph of the Methods section the following information to frame the rest of the methods:

“As this review was conducted to inform a WHO guideline, we followed the WHO Handbook for Guideline Development and used the recommended Grading of Recommendation, Assessment, Development, and Evaluation (GRADE) process to summarize findings.”

My initial thought on reading your response was to reluctantly agree to leaving the meta-analysis in the article, but it still doesn't sit well with me.

We appreciate this continued consideration of the meta-analysis. After substantial consideration, we decided to retain the meta-analysis overall, since that was what was originally presented to the WHO Guideline Development Group as part of the GRADE tables for this recommendation. However, have revised the presentation of findings to highlight the individual study results rather than the meta-analysis. In the Abstract, we have removed mention of the meta-analysis and instead summarize individual study results. In the Results section, we have added substantial additional detail on the findings of each of the 4 studies separately, before we present the meta-analysis. In the Discussion

section, we have added discussion of what the results are without the meta-analysis, and the limitations of the meta-analysis. We outline these changes in specific responses to your points below.

The concerns I have are:
1. Does the result of the meta-analysis add anything important to the GRADE table -- would you have arrived at a different recommendation if you hadn't done the meta-analysis?

We only had sufficient data to conduct a meta-analysis for 1 of the 7 outcomes that we originally specified *a priori.* Further, the included studies measure the outcome of "loss to follow-up" after abortion care, while our overall outcome category of interest was "appropriate clinical follow-up", which is quite different and which originally was intended to include follow-up for both pregnancy and antenatal care, contraceptive counseling and abortion care. This was discussed during the WHO guideline development committee meeting and there was little emphasis made to the particular outcome of loss-to-follow-up, particularly given that it was focused only on post-abortion care. So we do not think that an alternative way of presenting this would necessarily have changed the overall recommendation.

However, in response to this concern, we have revised the presentation of findings to highlight the individual study results rather than the meta-analysis. In the Abstract, we have removed mention of the meta-analysis and instead summarize individual study results. This sentence now reads as follows:

"For effectiveness, four randomized trials following 5493 individuals after medical abortion showed no difference or improvements in loss to follow-up with home pregnancy self-testing compared to return clinic visits."

In the Results section, we have added substantially more information about the individual study results in a new paragraph in the results section. This new paragraph reads as follows:

"One RCT from Vietnam found dramatically improved follow-up in the pregnancy self-testing arm, with only 0.6% of individuals lost compared with 8.1% lost in the comparison arm. The other three RCTs found no significant differences between study arms. In India, a study of 731 post-abortion participants from urban and rural Rajasthan found loss to follow-up was 3.7% in the self-testing arm compared to 4.8% in the clinic arm. In urban Austria, Finland, Norway, and Sweden, loss to follow-up among 929 post-abortion participants was 19.5% in the self-testing arm compared to 22.1% in the clinic arm. Finally, the largest study of 2400 post-abortion participants from Moldova and Uzbekistan found <1% in the self-testing arm compared to no loss to follow-up in the clinic arm."

To further clarify that this was only one part of the overall PICO category, we have now added to the GRADE Evidence Profile (Table 2) information about the category of outcome (from our original PICO question) that each measured outcome fits within.

In the discussion section, in the first sentence of the paragraph where we discuss the main findings from the effectiveness review, we have now highlighted the individual study results in addition to the overall meta-analysis result. This sentence now reads as follows:

"In the effectiveness review, we found low-certainty evidence from four RCTs following individuals after medical abortion showing no difference in loss to follow-up with self-testing for pregnancy at home compared to a return clinic visit, although individual study results varied, and one did show substantial improvements in loss to follow-up with pregnancy self-testing."

2. I briefly reviewed the Annex to the WHO guidelines with the methods from WHO and GRADE and didn't see a requirement to perform a meta-analysis in either (perhaps I missed it) -- could you explain your thinking about what it adds?

You are correct that there is not (to our knowledge) a requirement to conduct meta-analysis for the WHO and GRADE processes. However, GRADE is structured around presenting findings of each outcome (across studies) in a single row. This makes it difficult to present the results of multiple studies separately, and lends itself towards meta-analysis. The results of individual studies were

through the Forest Plot (Figure 2) and are now included in a new paragraph in the Results (described above). The meta-analysis simply summarizes (in an easier to understand way for the Guideline Development Group) what the overall assessment of this outcome is.

3. How would removing the meta-analysis affect the overall message and evidence of this article?

We believe the edits made above (detailed in our response to point 1) now nuance the overall message about the effectiveness; if anything, highlighting the one study that found a positive effect (rather than no difference) improves the strength of evidence. However, given the many limitations to this evidence (the fact that these are all from post-abortion studies and only one part of our *a priori* outcome of interest for decision-making), we have chosen not to change our statements about the overall message and evidence from the data available in the effectiveness review.

4. Computing power makes it easy to plug in numbers and produce quantitative results that appear authoritative but may not be.

We agree and have added the following sentence in the Discussion to highlight this point.

“While our meta-analysis summarizes the findings of the four effectiveness studies, it may provide a false sense of certainty that this represents the true effect.”

5. As an aside -- because I still think there are too few studies and too much heterogeneity to justify the meta-analysis -- you write that "Stratification by high- vs. low- and middle-income settings did not yield meaningful differences; no further stratifications were available." However, it seems that income category is just one of several potential variables you might stratify on.

Thank you. We had *a priori* listed the following variables we planned to stratify on (included in the Methods section):

“All analyses were stratified by the following categories/subgroups, where possible:

- Pregnancy self-test point of access (e.g. pharmacy, online, community health worker (CHW) distribution)
- Type of pregnancy test
- Population (e.g. age, marital status, rural/urban)
- Vulnerabilities (e.g. obesity, socio-economic status, poverty, disability, literacy/educational level)
- Gestational age
- High-income versus low or middle-income countries”

However, we ultimately identified only one outcome that was measured in multiple studies, and most of these categories/subgroups were not relevant (e.g., they were all post-abortion care studies so did not differ on point of access) or not reported (e.g. gestational age). The income category was the only factor that we felt might be relevant to assess, but with only four studies, it did not seem to yield a meaningful insight into the influence of country income on this outcome.

We have now edited the above section of the methods section to note that “We planned to stratify all analyses by the following categories/subgroups”. We hope this highlights that while we *planned* to do substantial subanalyses, we were unable to do so due to the nature of the data available. We have also changed the relevant sentence in the results section so that it now reads as follows:
“Stratification by high- vs. low- and middle-income settings did not yield meaningful differences; no further planned stratifications were possible, given the small number of similar studies identified for this outcome.”

If the meta-analysis stays in the article, I suggest adding more disclaimers and qualifiers about its applicability and interpretation than the brief sentences in the revised version.

In the discussion section, we have now edited the paragraph on limitations to highlight the following points:

“Importantly, effectiveness data was heavily skewed towards literature on post-abortion care, representing only a fraction of individuals using pregnancy self-tests globally.”

and

“Further, we only had sufficient studies to meta-analyze the outcome “lost to follow-up” after abortion, which provides just one specific way of assessing the overall outcome category of “appropriate clinical follow-up”.”

This is in addition to the sentence on the potential for a false sense of certainty noted in our response to point 4 above.

We hope these edits overall provide a better framing for the results on post-abortion follow-up and how it fits into the overall evidence base for pregnancy self-testing.

Reviewer: 1
Dr. Julie Shaw, Eastern Ontario Regional Laboratories Association Comments to the Author:
I still have concerns about the sensitivity and specificity information cited in the introduction for home pregnancy tests. I am asking the authors to address the fact that not all tests detect the same forms of B-hCG, which will alter sensitivity and specificity. There are some good papers in the literature on this issue. (Gronowski 2009, Nerenz 2014).

Thank you for this comment and the suggested references. We have reviewed them and have now added them to the introduction along with the following sentence: “test performance varies based on characteristics of the test, such as what form of hCG is detected”.

It is also important to address that tests can be highly sensitive if the tests are performed as per the manufacturer's instructions. This is an important point given these are performed outside of the lab.

We have highlighted this point through edits to the second sentence of the introduction, which now reads as follows: “The current generation of tests have close to 100% sensitivity and specificity in detecting hCG at concentrations of 25 milli-IU/ml or more, and thus are able to detect pregnancy as early as one day after a missed menstrual period, if performed per the manufacturers’ instructions.”

Reviewer: 2
Dr. C. Christina Mehta, Emory University School of Medicine Comments to the Author:
The revised manuscript adequately addresses all previous comments/questions.

Thank you for your continued review.

VERSION 3 – REVIEW

REVIEWER	Julie Shaw Eastern Ontario Regional Laboratories Association
REVIEW RETURNED	31-Jan-2022

GENERAL COMMENTS	The authors have addressed the concerns I raised on first review of this manuscript. No additional concerns.

REVIEWER	Karen Cowgill University of Washington
REVIEW RETURNED	18-Jan-2022

GENERAL COMMENTS	Thank you for your thoughtful and thorough responses and revisions.